Matrix functions and stability

Complete Crouzeix conjecture

Difficulty: extreme **Importance:** broadly interesting

Status: Partially resolved

Topic: Matrix-valued polynomial functional calculus

Rating rationale: The uniform bound must survive arbitrary matrix amplification, beyond the scalar functional calculus; it would give sharp bounds for block matrix functions and connect numerical-range estimates to dilation and similarity theory.

Context and notation

All matrices are complex, and all norms are Euclidean operator norms. For $A \in \mathbb{C}^{n\times n}$ define

$$W(A) = \{x^*Ax : x \in \mathbb{C}^n,\ x^*x = 1\}.$$

If $F(z) = [f_{ij}(z)]_{i,j=1}^m$ has polynomial entries, let $F(A) = [f_{ij}(A)]_{i,j=1}^m$, an $mn \times mn$ block matrix.

Problem statement

Prove or disprove that, for every $n, m \geq 1$, every $A \in \mathbb{C}^{n\times n}$ and every matrix polynomial $F \in \mathbb{C}^{m\times m}[z]$,

$$\|F(A)\|_2 \leq 2 \max_{z\in W(A)} \|F(z)\|_2.$$

This is the complete, rather than scalar, numerical-range spectral-set question. It controls simultaneous polynomial functions and hence block approximation errors with one constant independent of matrix and block sizes.

References

- Michel Crouzeix, *Numerical range and functional calculus in Hilbert space*, *Journal of Functional Analysis* **244**(2), 668–690 (2007). Original source for the complete conjecture, as explicitly attributed in the next source.
- Per Åhag, Rafał Czyż and Jani Virtanen, *Square Functions, Complete Crouzeix Conjecture in Dimension Three, and the Clouâtre–Ostermann–Ransford conjecture*, arXiv:2608.27346v3, **9 September 2026**, Introduction, equation **(1.2)** and **Theorem 1.1**. This current version states the exact target and explicitly distinguishes it from the scalar proofs. It claims the complete bound for $n \leq 3$. Its counterexample reduction leaves only 2×2 matrix polynomials to check when $n = 4$.
- Michel Crouzeix and César Palencia, *The numerical range is a $(1+\sqrt{2})$-spectral set*, *SIAM Journal on Matrix Analysis and Applications* **38**(2), 649–655 (2017). The complete universal upper bound is $1+\sqrt{2}$.

Scope and status check

The target is source-stated, not an editorial extension of the scalar conjecture. The August 2026 scalar proof claims by Jin and by Lorist–Schwenninger do not settle it: the September 9 primary manuscript explicitly addresses this distinction. The $n = 3$ result is a recent preprint claim; the full displayed target remains unresolved even if that claim is accepted. Searches through the check date for complete Crouzeix proofs, matrix-valued Crouzeix inequalities and later versions found no full solution. The version was inspected as both HTML and PDF; the inequality is on PDF page 2, immediately before Theorem 1.1.